DELHI PUBLIC SCHOOL JANKIPURAM, LUCKNOW

SESSION 2025-2026: PERIODIC ASSESSMENT-II

Subject: Artificial Intelligence (417)

SET–A

TIME: 2 Hrs.

M.M.:50

Date: _____ **Name:** _____ **Class/ SecX**_____

General Instructions:

1. Please read the instructions carefully.
2. This Question Paper consists of **21 questions** in two sections: Section A & Section B.
3. Section A has Objective type questions whereas Section B contains Subjective type questions.
4. **Out of the given (5 + 16 =) 21 questions, a candidate has to answer (5 + 10 =) 15 questions in the allotted (maximum) time of 2 hours.**
5. All questions of a particular section must be attempted in the correct order.
6. **SECTION A - OBJECTIVE TYPE QUESTIONS (24 MARKS):**

i. This section has 05 questions.

ii. Marks allotted are mentioned against each question/part.

iii. There is no negative marking.

iv. Do as per the instructions given.

7. **SECTION B – SUBJECTIVE TYPE QUESTIONS (26 MARKS):**

i. This section has 16 questions.

ii. A candidate has to do 10 questions.

iii. Do as per the instructions given.

iv. Marks allotted are mentioned against each question/part.

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SECTION A: OBJECTIVE TYPE QUESTIONS		
Q1.	**Answer any 4 out of the given 6 questions on Employability Skills (1 x 4 = 4 marks)**	
i.	Sneha loved playing online games. She met a person in the gaming chat who behaved very friendly and offered her free gaming points in exchange for sharing her personal e – mail and some private pictures. Sneha left unconfortable and reported the person to her parents. Which ICT risk did Sneha encounter?	1
ii.	A _____ is a type of computer virus that is primarily design to infect a specific	1

	type of document such as word or excel files. (a) Program File Virus (b) Boot Sector Virus (c) Macro Virus (d) E – mail Virus	
iii.	**Assertion (A):** Risk bearing is an essential characteristic of an entrepreneur. **Reason (R):** Entrepreneurs work in uncertain environments where outcomes are not always predictable. a. Both A and R are true and R is the correct explanation of A. b. Both A and R are true but R is not the correct explanation of A. c. A is true but R is false. d. A is false but R is true.	1
iv.	Which quality of an entrepreneur helps him to persist through difficult time and continuously seek options to ensure their businesses survive and flourish? (a) Optimistic (b) Patient (c) Hard working (d) Perseverant	1
v.	Which goal ensures equal access to quality health care, reducing illness and deaths by using preventive measures from communicable and non – communicable diseases? (a) Goal 4 – "Quality Education" (b) Goal - 2 "Zero Hunger" (c) Goal 3 – "Good Health and well being" (d) Goal - 1 "No Poverty"	1
vi.	Learning _____ is one of the major ways to enhance knowledge, abilities, values and attitudes. (a) Good communication (b) Green Skills (c) Healthy habits (d) Interpersonal skills	1
Q2.	**Answer any 5 out of the given 6 questions (1 x 5 = 5 marks)**	
i.	**Assertion (A):** Evaluation of an AI model is important to check its accuracy and performance. **Reason (R):** Without evaluation, we can not know if the model is making correct prediction or not. a. Both A and R are true and R is the correct explanation of A. b. Both A and R are true but R is not the correct explanation of A. c. A is true but R is false. d. A is false but R is true.	1
ii.	The _____ data set is used to make the model learn how to recognise patterns and relationships in the data. (a) Measuring (b) Training (c) Testing (d) Duplicate	1
iii.	Statement 1: Model evaluation is the process of applying various matrices to access a machine learning model's performance. Statement 2: Model evaluation helps the model to identify its strength, pinpoint weaknesses and determine how well it fits the task at hand. (a) Both Statement1 and Statement2 are correct. (b) Both Statement1 and Statement2 are incorrect. (c) Statement1 is correct but Statement2 is incorrect. (d) Statement2 is correct but Statement1 is incorrect.	1
iv.	In overfitting, the model is too complex and performs well on _____ data but poor on _____ data. (a) Testing, training (b) Training, testing (c) Developed, undeveloped (d) Structured, unstructured	1
v.	The confusion matrix is also known as _____ matrix. (a) Diluted (b) Error (c) Saturated (d) Filtered	1
vi.	What should be the **False Positive** case of - **"a self driving car detects obstacles on the road"**? (a) Car correctly identifies a harmless shadow as an obstacle.	1

	(b) Car incorrectly identifies a harmless shadow as an obstacle. (c) Car could not identify any shadow on the road. (d) None of the above.	
Q3.	**Answer any 5 out of the given 6 questions (1 x 5 = 5 marks)**	
i.	Train – Test Split technique provides _____ of performance of the machine learning model in real world scenario. (a) Biased estimate (b) Filtered estimate (c) Unbiased estimate (d) Splitted estimate	1
ii.	_____ is the domain of AI that enables computer to see, observe and understand digital images or data. (a) Statistical data (b) Computer Vision (c) Natural Language Processing (d) All of these	1
iii.	Which of the following task is an example of computer vision? (a) Rescaling an image (b) Changing tones of an image (c) Correcting brightness level in an image (d) Object detection in image	1
iv.	In which of the following task, computer vision helps self driving cars? (a) Identifying objects (b) Monitoring the environment (c) Navigating routes (d) All of these	1
v.	Every coloured image that is stored can be split in to _____ different channels. (a) Three (b) Four (c) Five (d) Two	1
vi.	The pixels in a grey scale image are arranged in _____ grid. (a) 1D (b) 2D (c) 4D (d) 3D	1
Q4.	**Answer any 5 out of the given 6 questions (1 x 5 = 5 marks)**	
i.	A retail store installs cameras with computer vision to count how many people enter and leave the store. Which AI application is being used? (a) Image classification (b) Object detection (c) Sentiment analysis (d) Speech recognition	1
ii.	In a grey scale image, what does the darkest shade represent? (a) Total presence of colour (b) Zero value of pixel (c) Lightest shade of grey (d) Maximum pixel value	1
iii.	_____ refers to the complete set of words and their meaning in a particular language. (a) Syntax (b) Lexicon (c) Semantics (d) Pragmatics	1
iv.	The process of creating the most meaningful and relevant summary of voluminous texts from multiple sources is known as _____. (a) Autogenerated captions (b) Language translation (c) Automatic text summarization (d) Text classification	1
v.	Which application of NLP helps business gain insights into consumer opinion, perform competitive comparisons and make necessary adjustments? (a) Sentiment and emotion analysis (b) Voice assistants (c) Text classification (d) Autogenerated captions	1
vi.	_____ is the process of automatically organising text in to predefined categories based on content and context. (a) Language translation (b) Keyword extraction (c) Text classification (d) Voice assistants	1
Q5.	**Answer any 5 out of the given 6 questions (1 x 5 = 5 marks)**	
i.	In which stage of NLP, the meaning of the sentence is drawn according to the meaning of preceeding and succeeding sentence? (a) Semantic analysis (b) Discourse integration (c) Pragmatic analysis (d) Lexical analysis	1

ii.	_____ is a powerful tool that automatically identifies the most frequently and contextually significant words or phrase in a text. (a) Language translation (b) Keyword extraction (c) Voice assistants (d) Classification	1
iii.	Which of the following is the third stage of natural language processing? (a) Syntactic analysis (b) Semantic analysis (c) Lexical analysis (d) Discourse integration	1
iv.	The smallest unit of information in a digital image is known as_____. (a) Pixel (b) Resolution (c) Width (d) Beam	1
v.	Google translate app uses a technology called_____, which allows it to see text in the image. (a) Optical code reader (b) Optical combined reader (c) Optical character reader (d) Optical control reader	1
vi.	_____ is the process of dividing an image into smaller segments to identify objects in it. (a) Object detection (b) Instance segmentation (c) Classification (d) Localisation	1

SECTION B: SUBJECTIVE TYPE QUESTIONS

	Answer any 3 out of the given 5 questions on Employability Skills (2 x 3 = 6 marks) **Answer each question in 20 – 30 words.**	
Q6.	What is the difference between Batch processing and Time sharing operating system?	2
Q7.	Write any two important roles that society plays to boost entrepreneurship?	2
Q8.	What is Goal 12 in SDG? What does this goal ensure?	2
Q9.	Write any two importance of sustainable development.	2
Q10.	**"Entrepreneurs are born not made".** Discuss this Myth about entrepreneurship.	2
	Answer any 4 out of the given 6 questions in 20-30 words each. (2 x 4 = 8 marks)	
Q11.	What do you understand by Precision and Recall? Explain with their formula.	2
Q12.	Explain the tem - True Positive and False Negative.	2
Q13.	What do you understand by classification in computer vision?	2
Q14.	Write any two features of natural language.	2
Q15.	What do you understand by text classification? Explain	2
Q16.	Write two differences between computer vision and image processing.	2
	Answer any 3 out of the given 5 questions in 50– 80 words each (4 x 3 = 12 marks)	
Q17.	Explain the working of train – test – split technique of model evaluation? Also explain why this technique is needed.	4
Q18.	Explain the process of searching the content by Google on the basis of image.	4
Q19.	Explain any four applications of NLP with proper headings.	4
Q20.	What do you undertand by Accuracy and Error in the confusion matrix? Explain with suitable example.	4
Q21.	An AI model made the following predictions for Book Sales forecast.	4

Confusion Matrix		Prediction	
		Yes	No
Actual	Yes	22	47
	No	12	18

Calculate Accuracy, Precision, Recall and F1 score for the above given confusion matrix.

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